

Modal Numerosity as Algorithmic Inference

A consolidated, camera-ready document compiling the formal framework, core theorems, tail bounds, structural restrictions, code snippets, and a research/validation roadmap.

Abstract

We develop **modal numerosity**: a quantitative semantics for questions of the form “How many P -things could there have been?” by pushing a world-measure λ forward along a **count functional** $\mathcal{N}_P$. The key technical move is to constrain λ via **effectivity** and (optionally) **structural axioms** (factorization, additivity, replication), yielding a nontrivial, mathematically characterizable space of attainable count distributions. We prove: (i) closure of count distributions under computable pushforwards, (ii) **tail bounds** for toy ecological numerosity under explicit prefix-code-like priors (polylog tails for Elias–delta-style priors; polynomial tails for power priors), and (iii) a structural restriction theorem showing that factorization + additivity impose convolution constraints. The framework generates empirical predictions (e.g., compressibility effects, tail-decay signatures) testable against graded human modal judgments.

1. Formal Framework

1.1 Effectivity prerequisites

- A function $f: \mathbb{N} \rightarrow \mathbb{R}_{\geq 0}$ is **lower semicomputable** (l.s.c.) if there exists a computable $g: \mathbb{N} \times \mathbb{N} \rightarrow \mathbb{R}_{\geq 0}$ such that for each n , $g(n, k)$ is nondecreasing in k and $g(n, k) \uparrow f(n)$ as $k \rightarrow \infty$.
- A **semimeasure** on a countable set X is a function $m: X \rightarrow [0, 1]$ with $\sum_{x \in X} m(x) \leq 1$. If equality holds, it is a probability measure.

1.2 World space and laws

Let W be a countable set of worlds/descriptions equipped with an **effective presentation**: a fixed bijection $\nu: \mathbb{N} \rightarrow W$.

A **law** L is a probability mass function on W that is **lower semicomputable pointwise**: there exists a computable $h: \mathbb{N} \times \mathbb{N} \rightarrow \mathbb{R}_{\geq 0}$ such that

$$h(j, t) \uparrow L(\nu(j)) \quad \text{as } t \rightarrow \infty,$$

with $\sum_j L(\nu(j)) = 1$.

Let $\mathcal{M} = \{L_0, L_1, \dots\}$ be a computably enumerable family of laws.

Let $\pi: \mathbb{N} \rightarrow [0, 1]$ be a lower semicomputable prior over laws with $\sum_i \pi(i) \leq 1$. Let $\pi_s(i)$ be a computable rational approximation with $\pi_s(i) \uparrow \pi(i)$.

1.3 Mixture semimeasure

Define the mixture semimeasure λ on $\mathbb{W}$ by

$$\lambda(w) = \sum_{i \in \mathbb{N}} \pi(i) \cdot L_i(w).$$

Then $\sum_{w \in \mathbb{W}} \lambda(w) \leq 1$.

1.4 Count functionals and pushforward

Let $N: \mathbb{W} \rightarrow \mathbb{N} \cup \{\infty\}$ be computable relative to the presentation (i.e., $N(\nu(j))$ is computable from j). The induced **count distribution** (pushforward) is

$$\lambda_N(n) = \lambda(\{w \in \mathbb{W} : N(w) = n\}).$$

Define tail $\tau(n) = \sum_{k \geq n} \lambda_N(k)$.

2. The Restriction Theorem (Effectivity Closure)

Theorem 1 (Closure under computable pushforwards)

If N is computable, then λ_N is a **lower semicomputable semimeasure** on $\mathbb{N} \cup \{\infty\}$.

Proof (sketch with full effectivity bookkeeping). For each i, n , because L_i is l.s.c. and N is computable, the mass

$$L_i(N=n) := L_i(\{w : N(w) = n\})$$

is l.s.c. uniformly in i, n . Let $g(i, n, s) \uparrow L_i(N=n)$ be a computable rational approximation.

Define

$$G(n, s) = \sum_{i=0}^s \pi_s(i) \cdot g(i, n, s).$$

Then $G(n, s)$ is computable, nondecreasing in s , and $G(n, s) \uparrow \lambda_N(n)$ as $s \rightarrow \infty$. Hence λ_N is l.s.c. The semimeasure property follows from $\sum_n \lambda_N(n) = \sum_w \lambda(w) \leq 1$. ■

Corollary 2 (Non-attainability via non-l.s.c.)

If a semimeasure q on $\mathbb{N}$ is **not** lower semicomputable, then it cannot be represented as λ_N for any computable-mixture construction above.

Concrete example. Let $A \subseteq \mathbb{N}$ be non-recursively enumerable. Define

$$q(n) = c \cdot 2^{-n} \cdot \mathbf{1}_A(n),$$

with c chosen to make $\sum_n q(n) \leq 1$. If q were l.s.c., one could enumerate A by waiting until the lower approximation to $q(n)$ exceeds a rational threshold, contradicting non-r.e.-ness of A . ■

3. Language-Relative Invariance (Type vs Token)

Let $\mathcal{L}$ be a modal/first-order language used to describe worlds, and let $\mathcal{P}$ be $\mathcal{L}$ -definable without haecceitistic constants.

Lemma (Definable numerosity is isomorphism invariant)

If $w \cong_{\mathcal{L}} w'$ ($\mathcal{L}$ -isomorphism), then $N_{\mathcal{P}}(w) = N_{\mathcal{P}}(w')$.

Consequence. For qualitative numerosity questions phrased in haecceity-free $\mathcal{L}$, only the induced measure on $\mathcal{L}$ -types (isomorphism classes / elementary equivalence classes, as appropriate) can affect $\lambda_{N_{\mathcal{P}}}$. Token-multiplicity within a type is invisible unless $\mathcal{L}$ can name individuals rigidly.

4. Tail Bounds under Explicit Priors (Toy Ecology)

4.1 Ecology model

A law $L = (S, E, \eta)$ has:

$$- \text{ } S \text{ } S \text{ } S \text{ } S \text{ } - \text{ } S \text{ } S \text{ } S \text{ } S \text{ } - \text{ } S \text{ } S \text{ } S \text{ } S \text{ }$$

Define capacity

$$N_{\max}(L) = \min(S^2, \lfloor E/\eta \rfloor).$$

Within-law models:

- **Model B (capacity):** $N = N_{\max}(L)$
- **Model A (uniform):** $N \mid L \sim \text{Unif}\{0, \dots, N_{\max}(L)\}$

Prior factorizes:

$$\pi(L) = \pi_S(S), \pi_E(E), \pi_{\eta}(\eta).$$

Define tail $\tau(n) = \Pr(N \geq n)$ under the mixture.

4.2 Dominance and factorization

Lemma 3 (Dominance). $\tau_A(n) \leq \Pr(N_{\max} \geq n) = \tau_B(n)$.

Lemma 4 (Factorization with η).

$$\Pr(N_{\max} \geq n) \leq \Pr\left(\bigvee_{\eta=1}^{\infty} S \geq \lceil \sqrt{n} \rceil\right) \cdot \sum_{\eta=1}^{\infty} \pi_{\eta}(\eta),$$

$$\Pr(E \geq n \mid \eta) = \Pr(E \geq n \mid \eta).$$

Corollary 5 (coarse bound). Since $\Pr(E \geq n \mid \eta) \leq \Pr(E \geq n)$,

$$\Pr(N_{\max} \geq n) \leq \Pr\left(\bigvee_{\eta=1}^{\infty} S \geq \lceil \sqrt{n} \rceil\right) \cdot \Pr(E \geq n).$$

4.3 Elias–delta-style priors and polylog tails

Assume for $x \geq 3$:

$$\pi_S(x) \leq \frac{C_S}{x(\log x)^2}, \quad \pi_E(x) \leq \frac{C_E}{x(\log x)^2}.$$

(Here $\log$ is natural log; changing base only changes constants.)

Lemma 6 (tail estimate). For $m \geq 3$,

$$\sum_{x \geq m} \frac{1}{x(\log x)^2} \leq \frac{1}{\log(m-1)}.$$

Theorem 7 (polylog tail bound; corrected constants). For $n \geq 16$,

$$\tau_B(n) \leq \frac{C}{(\log n)^2} \quad \text{with } C = 8C_{SC_E}.$$

Consequently $\tau_A(n) \leq C/(\log n)^2$.

Proof (sketch). From Cor. 5 and Lemma 6:

- For $n \geq 16$, $\Pr(S \geq \lceil \sqrt{n} \rceil) \leq 4C_S/\log n$.
- For $n \geq 4$, $\Pr(E \geq n) \leq 2C_E/\log n$. Multiply to get $\leq 8C_{SC_E}/(\log n)^2$. Use Lemma 3 for Model A. ■

4.4 Coding sensitivity: polynomial priors

Theorem 8 (power priors imply polynomial tails). If $\pi_S(x) \leq C_S/x^{\alpha}$ and $\pi_E(x) \leq C_E/x^{\alpha}$ for some $\alpha > 1$, then

$$\tau_B(n) = O(n^{\frac{3}{2}(\alpha-1)}).$$

(And $\tau_A(n) \leq \tau_B(n)$.)

5. Structural Restrictions: Factorization $\Rightarrow$ Convolution Constraints

5.1 Convolution framework

Let $(W, \otimes)$ be a commutative monoid modeling composition of independent subsystems.

Assume:

1. **Additivity:** $N(w \otimes w') = N(w) + N(w')$.
2. **Factorization (within a law):** for independent measurable sets A, B ,
3. **Replication closure:** for each L and $k \geq 1$, there is $L^{(k)}$ whose induced measure is the pushforward of $\mathbb{P}_{L^{\otimes k}}$ along $\otimes: W^k \rightarrow W$.

Let $p_L = (N)_* \mathbb{P}_L$ be the count distribution under L .

Theorem 9 (Convolution constraint)

$$p_{L^{(k)}} = p_L^{*k} \quad \text{({k-fold convolution})}$$

Thus any mixture $\lambda_N = \sum_L \pi(L) p_L$ lies in the convex hull of a convolution-closed family.

Corollary 10 (Structural non-attainability)

Any target distribution on $\mathbb{N}$ that cannot arise as a convex mixture of distributions obeying the convolution/replication constraints is unattainable under these structural axioms.

Interpretation. This is where “structure” bites harder than effectivity alone: effectivity permits all l.s.c. semimeasures in principle (via degenerate laws), but factorization + additivity + replication exclude many such constructions.

6. Attainability Landscape

Theorem 3.1 (Unrestricted attainability = all l.s.c. semimeasures)

If $\mathcal{M}$ is unrestricted enough to include point-mass (degenerate) laws L_n supported on a single designated world w_n with $N(w_n) = n$, then the attainable λ_N are **exactly** the l.s.c. semimeasures on $\mathbb{N}$.

- $(\Rightarrow)$ follows from Theorem 1.
- $(\Leftarrow)$ given any l.s.c. semimeasure q , set $\pi(n) = q(n)$ and use degenerate L_n .

Structural regime

With structural restrictions (e.g., Theorem 9's axioms), attainable distributions form a **proper subset**. Characterizing this subset under additional locality/invariance assumptions is a principal open problem.

7. Empirical Predictions and Validation

7.1 Predictions

1. **Tail-decay signatures** distinguish priors:
2. Elias–delta-style: $\tau(n) \lesssim (\log n)^{-2}$ (polylog)
3. Power priors: $\tau(n) \lesssim n^{-\beta}$ (polynomial)
4. Similarity/exponential: $\tau(n) \approx e^{-cn}$ or faster
5. **Compressibility spikes**: if numerosity is encoded directly in a simplicity-biased prior, “round” n (powers of 2/10, etc.) may receive elevated modal ease unless smoothed by the law/typicality layer.
6. **Moderator effects**: education/language may change the effective coding people use for numbers, modulating compressibility effects.

7.2 Minimal experiment sketch

- Prompt: “How easily could there have been at least n ravens?” (slider)
 - Use matched-magnitude pairs (e.g., 1024 vs 997; 1,000,000 vs 983,041)
 - Model ratings as monotone transform of $\tau(n)$ plus compressibility predictors
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8. Open Problems

1. **Exact structural characterization**: for given axioms (factorization/locality/invariance), characterize attainable λ_N .
 2. **Smoothing the prior**: formal conditions under which the law/typicality split removes numeral-encoding spikes.
 3. **Beyond discrete worlds**: extend to measurable spaces $\mathcal{W}$ via computable measure theory.
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Appendix A: Minimal Code Snippets

```
import numpy as np

def compute_tau_capacity(prior_S, prior_E, S_max=1000, E_max=1000, eta_max=1):
```

```

"""Compute  $\tau_B(n)=Pr(N\geq n)$  where  $N=N_{max}=\min(S^2, \text{floor}(E/\eta))$ .

This returns  $\tau[n] = Pr(N_{max} \geq n)$ . No extra tail transform is needed.
"""
maxN = min(S_max**2, E_max//1)
tau = np.zeros(maxN + 1, dtype=float)
total_w = 0.0

# simple factorized prior over S,E; optionally mix over eta
etas = np.arange(1, eta_max + 1)
w_eta = np.ones_like(etas, dtype=float) / len(etas)

for S in range(1, S_max + 1):
    wS = prior_S(S)
    S2 = S * S
    for E in range(1, E_max + 1):
        wE = prior_E(E)
        wSE = wS * wE
        for eta, weta in zip(etas, w_eta):
            Nmax = min(S2, E // eta)
            w = wSE * weta
            tau[:Nmax + 1] += w
            total_w += w

tau /= total_w
return tau

```

```

import numpy as np

def compute_pmf_and_tail_capacity(prior_S, prior_E, S_max=1000, E_max=1000,
eta_max=1):
    maxN = min(S_max**2, E_max)
    pmf = np.zeros(maxN + 1, dtype=float)
    total_w = 0.0

    etas = np.arange(1, eta_max + 1)
    w_eta = np.ones_like(etas, dtype=float) / len(etas)

    for S in range(1, S_max + 1):
        wS = prior_S(S)
        S2 = S * S
        for E in range(1, E_max + 1):
            wE = prior_E(E)
            wSE = wS * wE
            for eta, weta in zip(etas, w_eta):
                Nmax = min(S2, E // eta)

```

```

        w = wSE * weta
        pmf[Nmax] += w
        total_w += w

    pmf /= total_w
    tail = np.cumsum(pmf[::-1])[::-1]
    return pmf, tail

```

A.3 Example priors

```

import numpy as np

def elias_delta_style_prior(x, C=1.0):
    """Proxy for  $2^{-\ell_{\Delta}(x)}$  up to constants:  $\sim 1/(x (\log x)^2)$  for  $x \geq 3$ ."""
    if x < 3:
        return C / max(1, x)
    return C / (x * (np.log(x) ** 2))

def power_prior(x, alpha=2.0, C=1.0):
    return C / (x ** alpha)

```

Appendix B: External References (selected)

Algorithmic information / universal priors

- A. N. Kolmogorov — Foundations of algorithmic complexity (original papers; standard anthologies).
- R. J. Solomonoff — A formal theory of inductive inference (Parts I & II).
- L. A. Levin — Universal search / universal semimeasures.
- M. Li & P. Vitányi — *An Introduction to Kolmogorov Complexity and Its Applications*.
- M. Hutter — *Universal Artificial Intelligence: Sequential Decisions based on Algorithmic Probability*.

Prefix-free codes for integers

- P. Elias — Universal codeword sets and representations of the integers.

Computable analysis / computable measures

- K. Weihrauch — *Computable Analysis*.
- V. Brattka, P. Hertling, K. Weihrauch (eds.) — *Handbook of Computability and Complexity in Analysis*.

Modal semantics and measure

- D. Lewis — *On the Plurality of Worlds* (modal realism; similarity orderings; counterpart theory).
- Standard texts on probabilistic/graded modality in formal semantics and epistemology (for measure-on-worlds frameworks).

